



مدارس
للتنولوجيا
التطبيقية

IT Department
Web Development and Programming

تطوير المواقع والبرمجيات

الصف الثالث

Web Development and Programming
Combined question bank

2025 - 2026

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المعرفة	الكود
Writing asynchronous code using various techniques	TPK20

Dear Learner: Answer the following questions (7 questions).

1	Question	What is PHP mainly used for?
	Answer	Server-side web development.
2	Question	What is the purpose of Apache in PHP setup?
	Answer	It serves web pages to users.
3	Question	Write an example of a PHP variable.
	Answer	\$name = "Ali";
4	Question	How do you define a constant in PHP?
	Answer	define("PI", 3.14);
5	Question	What does the continue statement do in a loop?
	Answer	Skips current iteration.
6	Question	What is the difference between built-in and user-defined functions?
	Answer	Built-in functions are provided by PHP; user-defined functions are created by developers.
7	Question	What does the pow(2, 3) function return?
	Answer	8

Dear learner: Complete the question. (7 questions):

No	Question	Answer
8	PHP variables start with the symbol _____.	\$
9	PHP comments that span multiple lines use _____.	/* */
10	PHP constants are defined using the function _____.	define()
11	The default extension for PHP files is _____.	.php
12	To add two variables in PHP, we use the _____ operator.	+
13	The output function in PHP faster than print is _____.	echo
14	PHP scripts run on the _____ side.	Server

Dear Learner: Answer the following questions (8 questions).

15	Question	What is the main purpose of arrow functions?
	Answer	To write short, simple one-line functions.
16	Question	What is the output of ceil(4.2)?
	Answer	5
17	Question	What is the main difference between GET and POST?
	Answer	GET sends data in the URL, POST sends data in the request body (hidden).
18	Question	Why should sensitive data not be sent using GET?
	Answer	Because it appears in the URL and can be stored in browser history or logs.
19	Question	What is the purpose of server-side validation?
	Answer	To ensure data accuracy and security even if client-side validation is bypassed.
20	Question	What does htmlspecialchars() prevent?
	Answer	Cross-Site Scripting (XSS) attacks.
21	Question	What does CRUD stand for?
	Answer	Create, Read, Update, Delete.
22	Question	What is a row in a database table?
	Answer	A single record (entry) representing one set of related data.

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (7 questions).

No	Question	Answer
23	PHP is a client-side programming language.	False
24	PHP stands for "Hypertext Preprocessor."	True
25	PHP can be embedded inside HTML code.	True
26	Apache is a programming language used for building websites.	False
27	PHP scripts are executed on the server.	True
28	PHP variables must start with the \$ symbol.	True
29	Variable names in PHP are case-sensitive.	True

Give the Scientific Name Questions (8 questions).

No	Question	Answer
30	An array that stores values using numeric indexes is a.	Indexed Array
31	An array that uses named keys to store values is.	Associative Array
32	The function that counts array elements is.	count()
33	A PHP function that includes a file but continues execution even if missing.	include()
34	A PHP function that includes a file but throws a fatal error if missing.	require()
35	A PHP superglobal that collects form data sent via URL parameters.	\$_GET
36	A PHP superglobal that collects form data sent via HTTP POST.	\$_POST
37	A PHP statement that outputs content and is faster than print.	echo

Dear learner: Find the error in the code (6 questions).

No.	Question	Answer
38	<code><?php \$name = "Ali" echo \$name; ?></code>	Error: Missing semicolon after <code>`\$name = "Ali"``</code> Fix: <code>`\$name = "Ali";`</code>
39	<code><?php \$lname = "Ali"; echo \$lname; ?></code>	Error: Variable name cannot start with a number Fix: <code>\$lname1 = "Ali";</code>
40	<code>function greet(\$name) { echo "Hello " \$name; }</code>	Error: Missing dot (.) for concatenation. Fix: <code>echo "Hello " . \$name;</code>
41	<code>\$nums = array("1","2","3"); echo array_pop(\$nums, "4");</code>	Error: <code>array_pop()</code> takes only one parameter . Fix: <code>echo array_pop(\$nums);</code>
42	<code>echo strlen[“Hello”];</code>	Error: Wrong brackets . Fix: <code>echo strlen("Hello");</code>

43	<code>\$arr = array("name"=>"Ali", "age"=>20); echo \$arr[0];</code>	Error: Associative arrays require keys , not indexes. Fix: <code>echo \$arr["name"];</code>
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Dear learner: Choose the correct answer (7 questions).

No	Question	Answer
44	What does PHP stand for? a) Personal Hosting Program b) Hypertext Preprocessor c) Programming Hypertext Pages d) Page Hosting Processor	b
45	PHP code is written inside: a) <php> tags b) <script> tags c) <?php ?> tags d) {} braces	c
46	Which symbol is used to declare variables in PHP? a) @ b) \$ c) % d) &	b
47	Which statement is faster for output? a) echo b) print c) console.log d) display	a
48	Which operator is used for string concatenation in PHP? a) + b) & c) . d) ,	c
49	Single-line comments in PHP use: a) <!-- --> b) //	b

	c) \$ d) /* */	
50	Which operator assigns values to variables? a) Arithmetic b) Assignment c) Comparison d) Logical	b

المعرفة	الكود
Basic routing and calling a controller method from a route	TPK21

Dear Learner: Answer the following questions. (10 questions).

61	Question	What is ReactJS?
	Answer	A JavaScript library for building user interfaces using reusable components.
62	Question	What is JSX?
	Answer	A syntax extension for JavaScript that allows HTML-like code in JS files.
63	Question	What is the Virtual DOM?
	Answer	A lightweight copy of the real DOM for efficient updates.
64	Question	How does React handle rendering?
	Answer	It updates only the parts of the DOM that change (via Virtual DOM diffing).
65	Question	What are React components?
	Answer	Reusable UI building blocks (like custom HTML elements).
66	Question	What are props?
	Answer	Immutable data passed from parent to child components.
67	Question	How do you pass props to a component?
	Answer	<code><ChildComponent name="Yosra" age={30} /></code> .

68	Question	What is Vite?
	Answer	A fast build tool for modern React apps.
69	Question	How do you create a React app with Vite?
	Answer	Run npm create vite@latest and select React.
70	Question	How do you style React components?
	Answer	Inline styles, CSS modules, or libraries like Tailwind.

Dear learner: Complete the question (10 questions).

No	Question	Answer
61	JSX allows you to write _____-like code within JavaScript.	HTML
62	To manage state in functional components, you use the _____ Hook.	useState
63	Props in React are _____ (cannot be modified by child components).	Immutable
64	The tool recommended for creating modern React apps is _____.	Vite
65	For complex state logic, the _____ hook is preferred over useState.	useReducer
66	Bootstrap can be added to React via CDN or _____ manager.	Package
67	Tailwind CSS is configured in Vite through the _____ file.	vite.config.js
68	Inline styles in React are written as _____ objects.	JavaScript
69	Class components manage state using this. _____.	state
70	To share global data without prop drilling, you use the _____ hook.	useContext

Dear Learner: Answer the following questions (5 questions).

71	Question	What is React Router?
	Answer	A library for client-side navigation in SPAs.
72	Question	How do you define a route in React Router?
	Answer	<Route path="/about" element={<About />} />.

73	Question	What is <Link> used for?
	Answer	Client-side navigation without page reloads.
74	Question	How do you access URL parameters?
	Answer	const { id } = useParams();.
75	Question	How do you install Bootstrap in React?
	Answer	npm install bootstrap and import its CSS.

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

NO	Question	Answer
76	You can embed JavaScript expressions in JSX using curly braces {}.	True
77	JSX fragments are written as <...>.	True
78	React components must always start with a capital letter.	True
79	Functional components can't use state.	False
80	Class components are the modern preferred way to write components.	False
81	Components can return other components.	True
82	Props are immutable within child components.	True
83	State can only be used in class components.	False
84	setState() is asynchronous in class components.	True
85	useState returns the current state and a setter function.	True

Dear learner: Give signific term question (8 questions).

NO	Question	Answer
86	A utility-first CSS framework for rapid UI development.	Tailwind CSS
87	The library that enables client-side navigation in React applications.	React Router
88	The top-level component that syncs your app with the browser's URL.	BrowserRouter

89	The component that defines the mapping between URLs and components.	Route
90	The container component that holds multiple Route definitions.	Routes
91	The component used for navigation that prevents full page reloads.	Link
92	The special Link component that applies styling when active.	NavLink
93	The hook that extracts dynamic parameters from the URL.	useParams

Dear Learner: Answer the following questions (7 questions).

94	Question	What is state in React?
	Answer	Data that changes over time and affects rendering.
95	Question	How do you declare state in functional components?
	Answer	const [count, setCount] = useState(0);.
96	Question	What is useReducer used for?
	Answer	Managing complex state logic with actions/reducers.
97	Question	What is the purpose of useEffect?
	Answer	To perform side effects (e.g., API calls) in components.
98	Question	What is useContext?
	Answer	A hook for accessing context values without prop drilling.
99	Question	What is prop drilling?
	Answer	Passing props through multiple layers of components.
100	Question	What is Tailwind CSS?
	Answer	A utility-first CSS framework for rapid UI development.

المعرفة	الكود
Analyze and solve common web applications tasks by writing PHP programs	TPK 22

Dear Learner: Answer the following questions. (10 questions).

101	Question	What is Laravel?
	Answer	A free, open-source PHP framework for building web applications.
102	Question	What does MVC stand for in Laravel?
	Answer	Model, View, Controller.
103	Question	What command starts a Laravel development server?
	Answer	<code>php artisan serve.</code>
104	Question	Where are Laravel routes defined?
	Answer	In <code>routes/web.php</code> (for web) and <code>routes/api.php</code> (for APIs).
105	Question	What is Blade in Laravel?
	Answer	Laravel's templating engine for dynamic HTML rendering.
106	Question	What is Eloquent ORM?
	Answer	Laravel's database interaction system using PHP models instead of SQL.
107	Question	What is a migration in Laravel?
	Answer	A file that defines database table changes (create, modify, delete).
108	Question	How do you create a new controller?
	Answer	<code>php artisan make:controller ControllerName.</code>
109	Question	What is middleware in Laravel?
	Answer	A filter that checks/modifies HTTP requests before they reach routes/controllers.
110	Question	What is CSRF protection in Laravel?
	Answer	Security feature preventing cross-site request forgery attacks.

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
111	Laravel follows the MVC architecture.	True
112	MVC stands for Model, View, Controller.	True
113	Laravel does not enforce a separation between business logic and presentation.	False
114	Models in Laravel handle database interactions.	True
115	Views in Laravel are responsible for business logic.	False
116	Controllers act as an intermediary between Models and Views.	True
117	The default Laravel welcome page runs on http://localhost:8000.	True
118	Blade templates are stored in resources/views/.	True
119	The routes/ folder defines API and web routes.	True
120	Route parameters are defined using {} in Laravel.	True

Dear learner: Choose the correct answer (8 questions).

No	Question	Answer
121	What is Laravel? a) A JavaScript framework b) A PHP framework c) A CSS library d) A database system	b
122	Which architecture does Laravel follow? a) MVP b) MVC c) MVVM d) Flux	b
123	What does Laravel's Eloquent ORM do? a) Handles HTTP requests b) Compiles CSS c) Manages database interactions d) Renders Blade templates	c
124	Which tool is used to manage Laravel dependencies? a) npm	b

	b) Composer c) Yarn d) Pip	
125	What command starts the Laravel development server? a) php artisan start b) php artisan serve c) npm run dev d) composer serve	b
126	What is the purpose of middleware? a) Style web pages b) Filter HTTP requests c) Seed databases d) Generate fake data	b
127	Which method returns a view in Laravel? a) return view() b) return html() c) return blade() d) return render()	a
128	What does Route::resource() generate? a) A single route b) CRUD routes c) API endpoints d) Authentication routes	b

Dear Learner: Answer the following questions. (7 questions).

129	Question	What is the purpose of <code>php artisan make:model</code>?
	Answer	Creates a new Eloquent model linked to a database table.
130	Question	What is Laravel Breeze?
	Answer	A lightweight authentication package using Tailwind CSS.
131	Question	What is the difference between <code>PUT</code> and <code>PATCH</code>?
	Answer	<code>PUT</code> replaces entire resource, <code>PATCH</code> updates partial data.
132	Question	How do you generate fake data in Laravel?
	Answer	Using model factories and seeders.

133	Question	What is the purpose of @yield in Blade?
	Answer	Defines a placeholder for content in a layout.
134	Question	What is the file extension for Laravel's Blade templates?
	Answer	.blade.php
135	Question	What are the four basic operations of CRUD?
	Answer	Create, Read, Update, Delete.

Give the Scientific Name Questions (8 questions):

No	Question	Answer
136	What design pattern does Laravel use to separate logic, UI, and processes?	MVC
137	Which Laravel tool helps manage packages and dependencies?	Composer
138	What is the name of Laravel's command-line tool for generating code and managing tasks?	Artisan
139	What is the syntax used in Blade to output a variable safely?	{{ }}
140	What is the term for a gatekeeper that checks or modifies requests before they reach a route?	Middleware
141	What is the term for version-controlled files that modify database tables?	Migrations
142	What is Laravel's templating engine called?	Blade
143	What is the process of defining user permissions called?	Authorization

Test Code Debugging Challenge (7 questions):

144	Question	What's wrong with this migration? Schema::create('products', function (Blueprint \$table) { \$table->id(); \$table->string('name'); \$table->timestamps(); \$table->decimal('price'); });
	Answer	Fields should be added before timestamps(): Schema::create('products', function (Blueprint \$table) { \$table->id(); \$table->string('name'); \$table->decimal('price'); // Before timestamps \$table->timestamps(); });
145	Question	Fix this route? Route::get('/user', 'UserController@show');
	Answer	Route::get('/user', [UserController::class, 'show']);
146	Question	Why won't this update work? <form action="/users/{ {{ \$user->id }}" method="POST"> @csrf <!-- form fields --> </form>
	Answer	Missing @method('PUT'): <form action="/users/{ {{ \$user->id }}" method="POST"> @csrf @method('PUT') <!-- Add this --> <!-- form fields --> </form>
147	Question	Fix this validation: \$request->validate(['email' => 'required email']);
	Answer	Add unique rule for new records: \$request->validate(['email' => 'required email unique:users,email]);

148	Question	Find and fix the error. @foreach(\$users as \$user) {{ \$user.name }} @endforeach
	Answer	Wrong property access syntax. Fixed: @foreach(\$users as \$user) {{ \$user->name }} // Use -> not . @endforeach

149	Question	Why is this form vulnerable? <form method="POST" action="/login"> <!-- form fields --> </form>
	Answer	Missing @csrf: <form method="POST" action="/login"> @csrf <!-- Add this --> <!-- form fields --> </form>

150	Question	Fix this redirect to send flash message: return redirect('dashboard');
	Answer	Add flash message: return redirect('dashboard')->with('success', 'User created!');

المعرفة	الكود
software development	TPK 11

Dear Learner: Answer the following questions. (10 questions):

151	Question	What is software testing?
	Answer	Ensuring a product works as required and is free of bugs.
152	Question	Why is testing necessary?
	Answer	To find bugs early, improve quality, reduce costs, and meet user expectations.
153	Question	What is the difference between verification and validation?

	Answer	Verification checks if the software meets requirements; validation checks if it meets user needs.
154	Question	What is the difference between testing and debugging?
	Answer	Testing finds bugs; debugging fixes them.
155	Question	What is defect clustering?
	Answer	Most bugs are found in a few specific areas of the software.
156	Question	What is the pesticide paradox in testing?
	Answer	Repeating the same tests stops finding new bugs; fresh tests are needed.
157	Question	Name two test activities in the STLC.
	Answer	Test planning and test execution.
158	Question	What is testware?
	Answer	All artifacts (documents, scripts, tools) produced during testing.
159	Question	What is the role of a test analyst?
	Answer	Designs test cases based on requirements.
160	Question	What is shift-left testing?
	Answer	Performing testing earlier in the development lifecycle.

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
161	Software testing ensures that a product is free of bugs and performs as required.	True
162	Early testing helps reduce costs by catching bugs before they escalate.	True
163	The pesticide paradox suggests that repeating the same tests always finds new bugs.	False

164	Testing is the same for all types of software (e.g., games vs. banking apps).	False
165	Regression testing ensures that new changes don't break existing functionality.	True
166	Unit testing checks how different software components work together.	False
167	Performance testing checks how fast and stable software is under different workloads.	True
168	Shift-Left Testing means delaying testing until the end of development.	False
169	A retrospective meeting focuses only on what went wrong in a project.	False
170	A test case includes steps, expected results, and actual results.	True

Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
171	What is the main goal of software testing? a) To write code b) To ensure software works as expected and is bug-free c) To design user interfaces d) To manage project budgets	b
172	Which of the following is NOT a benefit of software testing? a) Early bug detection b) Increased development costs c) Improved software quality d) Meeting user expectations	b
173	Verification in software testing checks: a) If the software meets user needs b) Only the GUI design c) If the software meets specified requirements d) Only performance issues	c
174	What is the difference between testing and debugging? a) Testing fixes bugs; debugging finds them b) Testing finds bugs; debugging fixes them	b

	c) Both are the same d) Testing is done after debugging	
175	Which principle states that most bugs are found in a few specific modules? a) Defect clustering b) Early testing c) Pesticide paradox d) Exhaustive testing	a
176	Which testing type checks individual units or components of code? a) Integration testing b) System testing c) Unit testing d) Acceptance testing	c
177	What does the "pesticide paradox" imply in testing? a) Repeating the same tests finds new bugs b) New tests are needed to find new bugs c) Bugs are like pests that never go away d) Automated tests are better than manual tests	b
178	Which phase of STLC involves creating test cases? a) Requirement Analysis b) Test Planning c) Test Case Design d) Test Execution	c
179	Which of the following is a non-functional test? a) Unit testing b) Performance testing c) Integration testing d) Regression testing	b
180	What is the purpose of regression testing? a) To test new features only b) To replace unit testing c) To test only the user interface d) To ensure new changes don't break existing functionality	d

Dear learner: Complete the question (6 questions).

No	Question	Answer
181	The process of ensuring software is free of bugs and works as expected is called _____.	Software Testing
182	Checking if the software meets specified requirements is known as _____.	Verification
183	A set of steps to verify a feature's functionality is called a _____.	Test Case
184	Testing individual units of code is known as _____.	Unit Testing
185	The model that delivers software in small, usable increments is the _____ model.	Incremental
186	The tool used to track and manage bugs is called a _____.	Bug Tracking Tool

المعرفة	الكود
software testing approaches (analytical, model based, methodical...etc)	TPK 12

Dear Learner: Answer the following questions. (10 questions):

187	Question	What is the main goal of a retrospective in Agile?
	Answer	To reflect on what went well and identify improvements.
188	Question	What is a test case?
	Answer	A set of steps, conditions, and expected results to verify a feature.
189	Question	What is the difference between a test case and a test scenario?
	Answer	A test case is detailed; a test scenario is broader (e.g., "Check login" vs. "Enter username & password").
190	Question	What is a <u>defect</u> in software testing?
	Answer	An error causing the software to behave incorrectly.
191	Question	Name two common types of bugs.

	Answer	Syntax errors and logic errors.
192	Question	What is a bug tracking tool?
	Answer	Software used to record, track, and manage defects (e.g., Jira).
193	Question	What is the Waterfall model?
	Answer	A sequential SDLC model where each phase must be completed before moving to the next.
194	Question	What is Agile testing?
	Answer	Testing in short iterations with continuous feedback and adaptation.
195	Question	What is regression testing?
	Answer	Re-running old tests to ensure new changes don't break existing features.
196	Question	What is the purpose of UAT (User Acceptance Testing)?
	Answer	To confirm the software meets business/user needs before release.

Dear learner: Complete the question (6 questions).

No	Question	Answer
197	Testing software without predefined test cases is called _____ testing.	Exploratory
198	_____ testing checks individual code components in isolation.	Unit
199	_____ testing ensures different systems work together.	System Integration
200	_____ testing is done by real users before the final release.	Beta
201	A _____ describes a feature from a user's perspective.	User Story
202	_____ criteria define when a user story is complete.	Acceptance

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (12 questions).

No	Question	Answer
203	The Agile model focuses on early and continuous delivery of software.	True
204	User Acceptance Testing (UAT) is performed by developers.	False
205	Unit testing checks how different modules work together.	False
206	System Integration Testing (SIT) ensures seamless communication between different systems.	True
207	Alpha testing is performed by real users in a production environment.	False
208	Black-box testing requires knowledge of the internal code structure.	False
209	Boundary Value Analysis (BVA) tests values at the edges of input ranges.	True
210	Decision table testing is used when there are multiple conditions and actions.	True
211	State transition testing is used when the system behaviour changes based on different states.	True
212	Checklist-based testing is best suited for regression testing.	True
213	Acceptance criteria define when a user story is considered complete.	True
214	Collaborative user story writing involves only developers, not testers.	False

Match each term with its correct definition (6 questions).

No	Term	No	Definition	Answer
215	Testing	A	Fixing identified defects in the code.	215→B
216	Debugging	B	Detecting defects in the software.	216→A
217	Verification	C	Ensuring the software meets user needs.	217→D

218	Validation	D	Checking if the software meets specified requirements.	218→C
219	Error	E	A mistake made by a developer.	219→E
220	Defect	F	A flaw in the software causing incorrect behavior.	220→F

المعرفة	الكود
The basics of agile testing And API testing	TPK 13

Dear Learner: Answer the following questions (10 questions).

221	Question	What is the main goal of Agile methodology?
	Answer	Deliver value quickly through iterative development and customer collaboration.
222	Question	Name two Agile frameworks.
	Answer	Scrum and Kanban.
223	Question	What is a Sprint in Agile?
	Answer	A short development cycle (1-4 weeks) where a set of tasks is completed.
224	Question	What is the purpose of a Daily Scrum?
	Answer	To provide quick updates on progress and blockers.
225	Question	What does a Product Owner do in Scrum?
	Answer	Prioritizes tasks and manages the product backlog.
226	Question	Why is API testing important?
	Answer	It ensures APIs function correctly, securely, and efficiently.
227	Question	Name two common data formats used in API testing.
	Answer	JSON and XML.
228	Question	What is the difference between unit and integration testing in APIs?

	Answer	Unit tests check individual components, while integration tests check API interactions.
229	Question	What tool is used for testing REST APIs?
	Answer	Postman or SoapUI.
230	Question	What does security testing in APIs involve?
	Answer	AIdentifying vulnerabilities like SQL injection or unauthorized access.

Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
231	What is the primary focus of Agile methodology? a) Strict documentation b) Long development cycles c) Avoiding all changes d) Customer satisfaction and flexibility	d
232	Which Agile framework uses fixed-length sprints? a) Scrum b) Kanban c) Extreme Programming (XP) d) Waterfall	a
233	What is the main purpose of a Sprint Retrospective? a) To demo new features b) To improve team processes c) To assign new tasks d) To write documentation	b
234	Which role removes obstacles in Scrum? a) Product Owner b) Scrum Master c) Developer d) Stakeholder	b
235	Kanban uses: a) A visual board (To Do, In Progress, Done) b) Fixed sprints	a

	c) Only written reports d) No task tracking	
236	What does load testing simulate? a) A single API request b) Only manual testing c) Multiple users accessing an API d) Database queries	c
237	Which data formats are used in API testing? a) HTML and CSS b) XML and JSON c) PDF and DOCX d) MP3 and MP4	b
238	What is the main benefit of early bug detection in API testing? a) Saves time and money b) Increases costs c) Delays releases d) Requires more documentation	a
239	What does integration testing check? a) Single API components b) How different APIs work together c) Only security flaws d) UI responsiveness	b
240	Which testing ensures an API meets its intended purpose? a) Performance testing b) Load testing c) Validation testing d) Security testing	c

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (10 questions).

No	Question	Answer
241	Agile testing starts only after development is complete.	False
242	Daily Scrums are optional in Agile teams.	False
243	Kanban uses fixed sprints like Scrum.	False

244	The Product Owner in Scrum sets task priorities.	True
245	API testing helps detect bugs early in development.	True
246	Security testing is not part of API testing.	False
247	Load testing simulates a single user accessing an API.	False
248	Unit testing checks how different APIs work together.	False
249	JSON and XML are common data formats in API testing.	True
250	API testing reduces costs by catching errors early.	True

المعرفة	الكود
Database and data manipulation and Nodejs framework	TPK19

Give the Scientific Name Questions: (7 questions).

No	Question	Answer
251	The part of a website that represents the browser on the computer is called the:	client
252	The server or place that contains the backend code is called the:	Server
253	The open-source, cross-platform runtime environment for executing JavaScript code is:	Node.js
254	The open-source JavaScript and Web Assembly engine developed by Google is called:	V8
255	A Node.js framework known for its simple design that lets developers create customized applications is:	Express.js
256	A library that facilitates real-time communication by allowing clients and servers to exchange messages instantly is:	Socket.io
257	A type of operation where a task must finish before the next one can start is called:	Blocking

Dear Learner: Answer the following questions : (7 questions).

258	Question	What are the three main parts of a website's architecture?
	Answer	The client, the server, and the database.

259	Question	Define: Node.js?
	Answer	Node.js is an open-source, cross-platform runtime environment for executing JavaScript code outside a web browser.

260	Question	What is the key difference between blocking and non-blocking operations?
	Answer	Answer: Blocking operations must be completed before the next task can start, while non-blocking operations can run independently without waiting.

261	Question	What command do you use to check if Node.js is installed correctly on your system?
	Answer	node -v

262	Question	What is the primary purpose of Express.js?
	Answer	To simplify the process of building web applications and APIs in Node.js.

263	Question	What is the role of middleware?
	Answer	To handle and process requests and responses between the client and the server.

264	Question	What is the key structural difference between SQL and NoSQL databases?
	Answer	SQL uses tables with fixed rows and columns, while NoSQL uses flexible documents or other models.

265	Question	What is the main benefit of using a template engine like EJS?
	Answer	It separates the presentation layer (HTML) from the application logic (JavaScript).

266	Question	What command is used to start an Express server with automatic restarts on file changes?
	Answer	nodemon server.js

267	Question	What is Crud?
	Answer	The set of operations (Create, Read, Update, Delete) used to manage data

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (12 questions).

No	Question	Answer
268	Node.js itself is a programming language.	False
269	The V8 engine was developed by Google.	True
270	Node.js is considered a framework like Express.js.	False
271	When you install Node.js, NPM is installed automatically.	True
272	The http module is an example of a third-party module.	False
273	TypeScript code is executed directly by a web browser's JavaScript engine.	False
274	The private access modifier allows a class property to be accessed from anywhere in your code.	False
275	Inheritance allows a subclass to inherit properties and methods from a superclass.	True
276	Middleware functions handle requests and responses between the client and server.	True
277	A PUT request is used specifically for deleting records in a REST API.	False
278	The app.use(express.json()) middleware is necessary to parse JSON data from the request body.	True
279	The req.body object is used to access data sent in the request's JSON body.	True

Dear learner: Choose the correct answer (12 questions).

No	Question	Answer
280	What is Node.js? a) A JavaScript framework b) A new programming language c) A runtime environment for executing JavaScript code d) A web browser	c
281	How would you describe a "Non-Blocking" operation? a) Tasks must complete in order, one after the other. b) Tasks can start and run independently without waiting. c) It prevents other operations from starting. d) It is another term for a synchronous operation.	b
282	How do you verify the version of Node.js installed on your system? a) node -v b) npm -v c) install node d) run node	a
283	The fs (file system) module in Node.js is an example of what? a) A Third-Party Module b) A Local Module c) A Core Module d) A Framework	c
284	Which command is used to install TypeScript globally using npm? A) npm install typescript B) npm add -g typescript C) node install typescript D) npm install -g typescript	d
285	What is the correct file extension for a TypeScript source file? A) .ts B) .txt C) .js D) .typescript	a

286	What is the primary purpose of Express.js? a) To style web pages b) To build server-side web applications and APIs c) To manage client-side state d) To create database schemas	b
287	Which command is used to initialize a new Node.js project and create a package.json file? a) npm start b) node init c) npm install express d) npm init	d
288	Which of the following is NOT a feature of Express.js mentioned in the text? a) Middleware b) Template Engines c) Virtual DOM d) Error Handling	c
289	How is a specific employee's ID typically accessed from the URL in an Express route like /employees/:id? a) req.query.id b) req.body.id c) req.params.id d) req.url.id	c
290	What does CRUD stand for? a) Create, Read, Update, Delete b) Call, Render, Update, Destroy c) Create, Render, Use, Deploy d) Calculate, Return, Use, Define	a
291	The main purpose of CORS is to: A) Prevent all access to a web server. B) Securely allow web pages to make requests to a different domain. C) Encrypt data stored in a database. D) Improve the performance of template rendering.	b

Give the Scientific Name Questions: (7 questions).

No	Question	Answer
292	A type of operation where tasks can start and continue independently of each other is called:	Non-blocking
293	The tool that comes with Node.js for managing packages is the:	npm
294	A language that builds on JavaScript by adding static typing is called:	TypeScript
295	Functions that handle requests and responses between the client and server are known as:	Middleware
296	The acronym for an interface that connects different applications is:	API
297	The security mechanism that prevents web pages from making requests to a different domain is called:	CORS
298	The set of operations (Create, Read, Update, Delete) used to manage data is known as:	CRUD

المعرفة	الكود
Digital tools options and uses	TPK 18

Dear Learner: Answer the following questions (4 questions).

299	Question	What are Creational Patterns?
	Answer	Creational patterns focus on flexible and controlled object creation.

300	Question	What are Structural Patterns?
	Answer	Structural patterns focus on organizing classes and objects into larger, maintainable structures.

301	Question	Explain the main purpose of Behavioral Design Patterns.
	Answer	Behavioral design patterns define how objects communicate, distribute responsibilities, and collaborate, improving flexibility and scalability of applications.

302	Question	Discuss the benefits of using the Template Method pattern in large projects.
	Answer	It reduces code duplication, promotes reusability, and enforces a consistent algorithm structure. It allows flexibility through subclass customization while keeping the base algorithm stable.

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement (12 questions).

No	Question	Answer
303	SOLID principles are used in many design patterns.	True
304	The Open-Closed Principle states that a class should be open for extension and closed for modification.	True
305	Behavioral design patterns focus on object creation.	False
306	Structural design patterns are about organizing classes and objects into larger structures.	True
307	A single responsibility principle suggests that a class should have only one reason to change.	True
308	Creational patterns focus on class instantiation and object creation.	True
309	Design patterns speed up development by providing proven Solutions.	True
310	Design patterns are compared to Lego bricks because both help in building complex structures from smaller reusable units.	True
311	Creational design patterns are mainly concerned with how objects interact with each other.	False
312	Factory Pattern is used to ensure only one instance of a class exists.	False
313	Some design patterns can be combined to solve more complex problems.	True
314	The Singleton pattern ensures multiple instances of a class can exist at the same time	False

Dear learner: Choose the correct answer (7 questions).

No.	Question	Answer
315	Which of the following is NOT one of the SOLID principles? a) Single Responsibility Principle b) Open-Closed Principle c) Dependency Creation Principle d) Liskov Substitution Principle	C
316	Which design pattern category deals with communication between objects? a) Structural b) Behavioral c) Creational d) Functional	B
317	The main goal of the Interface Segregation Principle is: a) To increase inheritance b) To split large interfaces into smaller ones c) To combine interfaces d) To remove interfaces completely	B
318	Which of the following is the main task of design patterns? a) Provide high-level guidelines b) Provide reusable solutions to common design problems. c) Focus only on database logic d) Eliminate OOP concepts	B
319	Which of the following is a Creational Design Pattern? a) Adapter b) Singleton c) Composite d) Proxy	B
320	The main goal of Creational design patterns is: a) Code execution speed b) Object creation flexibility	B

	c) Database optimization d) User interface design	
321	Structural patterns are mainly concerned with: a) Data storage b) Object creation c) Organizing relationships between objects d) Error handling	C

Give the Scientific Name Questions (6 questions).

No	Question	Answer
322	A set of five design principles that make object-oriented software more maintainable and extensible.	SOLID
323	A pattern that defines how objects communicate and distribute responsibilities.	Behavioral Design Pattern
324	A design pattern where the algorithm structure is defined in a base class with customizable steps.	Template Method Pattern
325	Ability to use existing code structure across different subclasses.	Reusability
326	Modifying specific steps while keeping the general algorithm intact.	Customization
327	Ability of code to adapt to changes in behavior.	Flexibility

Complete the Sentence Questions (10 questions).

No	Question	Answer
328	In design patterns classification, the two criteria are _____ and _____.	Purpose, Scope
329	The _____ Pattern allows you to define a family of algorithms and switch between them at runtime.	Strategy
330	A charger plug converter is an example of the _____ Pattern.	Adapter

331	The three main types of design patterns are _____, _____, and _____.	Creational, Structural, Behavioral
332	The Factory Pattern provides a common _____ for creating objects of different subclasses.	Interface
333	----- patterns focus on how objects are created .	Creational design
334	The ----- pattern helps to make incompatible interfaces compatible.	Adapter
335	----- pattern can improve performance by saving memory.	Singleton
336	----- patterns are concerned with relationships between objects.	Structural
337	The----- Method pattern provides a uniform interface for creating objects.	Factory

المعرفة	الكود
Python and web GUI frameworks	TPK 17

Dear Learner: Answer the following questions. (7 questions).

338	Question	What is a GUI and what does it replace?
	Answer	A GUI (Graphical User Interface) allows users to interact with electronic devices using visual elements like icons and buttons, replacing the need for complex command-line interfaces.
339	Question	What are three examples of common GUI elements?
	Answer	Common GUI elements include buttons, menus, and text boxes.
340	Question	Why is Tkinter considered a good choice for beginners in Python GUI development?
	Answer	Tkinter is simple to learn and comes pre-installed with Python, making it easy to start with GUI development.

341	Question	In Tkinter, what is the purpose of the root widget?
	Answer	The root widget is the main or parent window of a Tkinter application, and all other widgets must be created within it.

342	Question	What are two ways to load images into a Tkinter label?
	Answer	You can use the PhotoImage class to load GIF, PGM, PPM, and PNG images, or use the Pillow library for other image formats.

343	Question	Briefly describe the purpose of the <u>command</u> parameter in a Tkinter Button widget.
	Answer	The command parameter is used to specify the name of the Python function or method that should be called whenever the button is clicked.

344	Question	What is the primary function of the tkinter.messagebox module?
	Answer	Its primary function is to display ready-made pop-up message boxes to the user to provide information, warnings, errors, or ask for confirmation.

Put "True" in front of the correct statement and "False" in front of the incorrect statement (7 questions).

No	Question	Answer
345	A Graphical User Interface (GUI) requires users to have deep programming knowledge to interact with electronic devices.	False
346	Buttons are clickable elements that trigger actions within a GUI.	True
347	PyQt is considered the easiest Python GUI library for beginners because it comes installed with Python.	False
348	The geometry() method in Tkinter sets the window's starting size and position.	True
349	The pack() method is a geometry manager used to position widgets in a window.	True

350	Tkinter's default look and feel is considered modern and sleek.	False
351	Tkinter.ttk is a modern set of widgets that automatically matches the style of the user's operating system.	True

Dear learner: Choose the correct answer (10 questions).

No	Question	Answer
352	Which of the following is NOT a common GUI element mentioned in the text? a. Text boxes b. Windows c. Command lines d. Scroll bars	c
353	According to the text, which Python GUI library is best for creating simple, basic programs and is easy to start with? a. Kivy b. PyQt c. PyGTK d. Tkinter	d
354	What is the main advantage of using Tkinter.ttk over standard Tkinter? a. It is better for mobile apps. b. It provides more advanced widgets. c. It automatically matches the look of the user's operating system. d. It is easier to customize colors.	c
355	What is the purpose of the title() method in a Tkinter window? a. To set the size of the window. b. To change the text at the top of the window. c. To set the background color. d. To set the minimum size of the window.	b
356	What is the root widget in a Tkinter application? a. A type of button.	b

	b. The main window or parent window. c. A field for text input. d. An image file.	
357	Which Python GUI library is described as "The Mobile One" and is best for apps with touchscreens? a. Tkinter b. PyQt c. Kivy d. wxPython	c
358	The geometry() method string format widthxheight+X+Y defines what for a window? a. Width and height only. b. Width, height, and title. c. Width, height, and starting position from the top-left corner. d. Width, height, and minimum size.	c
359	What is the purpose of the pack() method when used on a widget in Tkinter? a. To set the widget's color. b. To change the widget's font. c. To run the app's main event loop. d. To place the widget on the current window.	d
360	What is the disadvantage of using Tkinter mentioned in the text? a. It does not work on macOS. b. It requires an external library to be installed. c. Its default look and feel are dated. d. It is not compatible with other Python tools.	c
361	Which Python GUI library is recommended for making complex, professional applications that need to look native on different operating systems? a. Tkinter b. Kivy c. PyQt d. Pygame	c

Dear learner: Give The Correct Scientific Term (7 questions).

No	Question	Answer
362	This is a Tkinter geometry manager used to pack or place a widget on the current window.	pack()
363	This is the main window of a Tkinter application, which must be created before any other windows.	Root widget
364	This is the method used to set the starting size and position of a Tkinter window.	geometry()
365	This is the standard Python library that provides a simple and intuitive way to create graphical user interfaces.	Tkinter
366	This is a Tkinter widget that can be used to display either text or images.	Label
367	This specific method is used to change settings of a Tkinter window, such as its background color.	configure()
368	This is the part of a Tkinter application that handles mouse and keyboard inputs and communicates with the operating system.	Main event loop

المعرفة	الكود
Data structures	TPK 23

Match the function on the left with its correct description on the right. (6 questions).

No	Function	No	Description	Answer
369	date()	A	Returns an associative array with date/time information.	369 → c

370	time()	B	Parses an English textual datetime into a timestamp.	370 → e
371	strptime()	C	Formats a local date and time into a specified string.	371 → b
372	getdate()	D	Validates a Gregorian date (e.g., checks for leap year).	372 → a
373	checkdate()	E	Returns the current Unix timestamp.	373 → d
374	__construct()	F	A special method called when a new object is created.	374 → f

Give the Scientific Name Questions: (10 questions).

NO	Question	Answer
375	A function that is automatically called when you create an object from a class.	constructor
376	The keyword used to refer to the current object inside a class method.	\$this
377	The access modifier that allows a property to be accessed from anywhere in the code.	public
378	The OOP principle that helps keep code DRY by allowing classes to acquire properties from other classes.	inheritance
379	The programming paradigm that is about creating objects containing both data and functions.	OOP
380	A function that is automatically called at the end of a script or when an object is destroyed.	destructor
381	The principle of restricting direct access to some of an object's components.	encapsulation
382	An instance of a class.	object
383	A template for objects, defining properties and methods.	class
384	The principle that allows objects of different types to be treated as objects of a common supertype.	polymorphism

Complete the Sentence Questions: (8 questions).

No	Question	Answer
385	The <code>date()</code> function is used to format a local and time.	date
386	The <code>time()</code> function returns the current Unix timestamp.	time()
387	To create a new object, you use the <code>new</code> keyword.	new
388	The <code>this</code> keyword refers to the current object inside a class method.	\$this
389	A constructor function starts with two underscores: <code>__construct</code> .	__construct
390	A <code>private</code> property can only be accessed within the class itself.	private
391	The OOP principle of <code>abstraction</code> focuses on hiding complex details and showing only essential features.	abstraction
392	In OOP, a class is a <code>template</code> for objects.	template

Dear learner: Answer the following questions. (8 questions).

393	Question	What does OOP stand for and name one advantage it has over procedural programming?
	Answer	Object-Oriented Programming. Advantages include: clearer structure, easier to maintain/modify/debug, code reuse (DRY), faster development.

394	Question	What is the difference between a class and an object?
	Answer	A class is a template or blueprint. An object is an instance of a class.

395	Question	What is the primary purpose of a constructor method (<code>__construct</code>)?
-----	-----------------	--

	Answer	To initialize an object's properties automatically when the object is created.
--	---------------	--

396	Question	What does the acronym DRY mean in the context of OOP advantages?
	Answer	Don't Repeat Yourself. It means code is reusable and not duplicated.

397	Question	Name the three access modifiers in PHP OOP.
	Answer	public, protected, private.

398	Question	What is the main benefit of the Inheritance principle?
	Answer	It promotes code reuse by allowing a new class to inherit properties and methods from an existing class.

399	Question	What does the \$this keyword represent inside a class method?
	Answer	It represents the current object instance calling the method.

400	Question	What are the two essential attributes an HTML form must have to allow file uploads?
	Answer	method="POST" and enctype="multipart/form-data".

المعرفة	الكود
Algorithms	TPK24

Dear Learner: Answer the following questions. (4 questions).

401	Question	What is the primary technical difference between TypeScript and JavaScript?
	Answer	TypeScript is statically typed, while JavaScript is dynamically typed.

402	Question	What is the purpose of the tsconfig.json file?
	Answer	To configure the behavior of the TypeScript compiler (defines compiler settings).

403	Question	What command do you use to compile a TypeScript file named app.ts?
	Answer	tsc app.ts

404	Question	What are the four main principles of Object-Oriented Programming supported by TypeScript?
	Answer	Encapsulation, Inheritance, Polymorphism, Abstraction.

Dear learner: Put "True" in front of the correct statement and "False" in front of the incorrect statement. (8 questions).

No	Question	Answer
405	File handling in PHP can only be used to read files, not write to them.	False
406	The enctype attribute for a file upload form must be set to multipart/form-data.	True
407	The \$_FILES superglobal array is used to access information about an uploaded file.	True
408	The move_uploaded_file() function is used to delete a file from the server.	False
409	PDO stands for PHP Data Orientation.	False

410	PDO can only be used to connect to MySQL databases.	False
411	Prepared statements in PDO help prevent SQL injection attacks.	True
412	Cookies are stored on the server.	False

Complete the Sentence Questions: (10 questions).

No	Question	Answer
413	The form attribute enctype must be set to _____ for file uploads to work.	multipart/form-data
414	PDO is an abbreviation for _____.	PHP Data Objects
415	To connect to a database with PDO, you need to create a new PDO instance with a connection string known as a _____.	DSN (Data Source Name)
416	The _____ method of a PDO object is used to create a prepared statement. Answer:	prepare
417	To set a cookie to expire in one day, you would set the expire parameter to time() + _____ seconds.	86400
418	You should use the _____ function to check if a cookie is set before trying to use its value.	isset
419	Session variables are stored on the _____, while cookies are stored on the _____.	server, client
420	The server identifies a user's session on subsequent requests using a unique _____ that is typically stored in a cookie.	session ID
421	In the PDO DSN for MySQL, you specify the database driver using _____ :.	mysql

422	The _____ method of a PDO statement object is used to fetch a single row from a result set.	fetch
-----	---	-------

Dear learner: Choose the correct answer (8 questions).

No	Question	Answer
423	Which function is crucial for moving an uploaded file to a permanent location? a) copy_file() b) move_uploaded_file() c) save_upload() d) store_file()	b
424	Which HTML form attribute is essential for enabling file uploads? a) method="GET" b) action="upload.php" c) enctype="multipart/form-data" d) type="file"	c
425	Where is information about an uploaded file temporarily stored before being processed by PHP? a) \$_FILES["file"]["tmp_name"] b) \$_FILES["file"]["perm_name"] c) \$_POST["file"] d) \$_GET["file"]	a
426	What is the primary advantage of using PDO over the older mysql extension? a) It's faster b) It provides a consistent interface for multiple databases c) It doesn't require a password d) It only works with MySQL	b
427	Which method is used to execute a prepared statement in PDO? a) run() b) execute()	b

	c) query() d) fetch()	
428	Which function is used to create a cookie in PHP? a) create_cookie() b) make_cookie() c) setcookie() d) set_cookie()	c
429	Where is session data stored by default? a) On the client's browser b) In a database c) In a cookie d) On the server	d
430	Which function must always be called before using \$_SESSION variables? a) session_begin() b) start_session() c) session_start() d) session_init()	c

المعرفة	الكود
Consol app	TPK25

Dear learner: Answer the following questions. (5 questions).

	Question	Name two major advantages of using PDO over the older mysql_i extension.
431	Answer	1. works with many databases, not just MySQL. 2. Strong, consistent support for prepared statements to prevent SQL injection.

432	Question	Briefly describe the difference between session_unset() and session_destroy().
	Answer	session_unset() removes all session variables, while session_destroy() destroys the

433	Question	What is the key difference between where cookies and sessions store their data?
	Answer	Cookies are stored on the client's browser, while sessions are stored on the server.

434	Question	What is the purpose of the session ID?
	Answer	It is a unique identifier stored on the client.

435	Question	How does session_start() work with cookies?
	Answer	It sends a Set-Cookie header with the session ID (PHPSESSID) to the client browser to maintain the session.

Dear learner: Give the Correct Scientific Term (7 questions):

No	Question	Answer
436	This geometry manager places each widget in its own box and can stack them vertically or horizontally by default.	pack
437	This argument for the pack manager defines where the widget is placed in the parent widget, such as "n" for North or "s" for South.	anchor
438	This geometry manager arranges widgets within a matrix of rows and columns, like a spreadsheet.	grid

439	This geometry manager allows for absolute control over a widget's position using specific x and y coordinates.	place
440	This Tkinter widget allows you to display and edit multi-line text and supports embedded images and links.	Text widget
441	This method is used to add content to a Text widget, taking an index and the text characters as arguments.	insert()
442	The index format used by the Text widget's get() and insert() methods, which follows a 'line.column' structure.	index

Dear learner: Complete the question (8 questions):

No	Question	Answer
443	User interfaces are made up of multiple _____, arranged within a window.	widgets
444	You can set the text of a Label widget by using the _____ function.	config()
445	To align Label text to the left edge, you would set the anchor option to _____ or tk.W.	"w"
446	The _____ widget allows users to type input and can handle events like the Return key being pressed.	Entry
447	The actions which buttons perform are handled by Python _____.	functions
448	The root._____() function is called to immediately close the application window.	destroy
449	The bd parameter of a Button sets the _____ width in pixels.	border
450	The askokcancel and askyesno functions are types of _____ boxes that ask for user confirmation.	message

TPK20	TPK21	TPK22	TPK11	TPK12	TPK13	TPK19	TPK18	TPK17	TPK23	TPK24	TPK25
1	51	101	151	187	221	251	299	338	369	401	431
2	52	102	152	188	222	252	300	339	370	402	432
3	53	103	153	189	223	253	301	340	371	403	433
4	54	104	154	190	224	254	302	341	372	404	434
5	55	105	155	191	225	255	303	342	373	405	435
6	56	106	156	192	226	256	304	343	374	406	436
7	57	107	157	193	227	257	305	344	375	407	437
8	58	108	158	194	228	258	306	345	376	408	438
9	59	109	159	195	229	259	307	346	377	409	439
10	60	110	160	196	230	260	308	347	378	410	440
11	61	111	161	197	231	261	309	348	379	411	441
12	62	112	162	198	232	262	310	349	380	412	442
13	63	113	163	199	233	263	311	350	381	413	443
14	64	114	164	200	234	264	312	351	382	414	444
15	65	115	165	201	235	265	313	352	383	415	445
16	66	116	166	202	236	266	314	353	384	416	446
17	67	117	167	203	237	267	315	354	385	417	447
18	68	118	168	204	238	268	316	355	386	418	448
19	69	119	169	205	239	269	317	356	387	419	449
20	70	120	170	206	240	270	318	357	388	420	450
21	71	121	171	207	241	271	319	358	389	421	
22	72	122	172	208	242	272	320	359	390	422	
23	73	123	173	209	243	273	321	360	391	423	
24	74	124	174	210	244	274	322	361	392	424	
25	75	125	175	211	245	275	323	362	393	425	
26	76	126	176	212	246	276	324	363	394	426	

27	77	127	177	213	247	277	325	364	395	427	
28	78	128	178	214	248	278	326	365	396	428	
29	79	129	179	215	249	279	327	366	397	429	
30	80	130	180	216	250	280	328	367	398	430	
31	81	131	181	217		281	329	368	399		
32	82	132	182	218		282	330		400		
33	83	133	183	219		283	331				
34	84	134	184	220		284	332				
35	85	135	185			285	333				
36	86	136	186			286	334				
37	87	137				287	335				
38	88	138				288	336				
39	89	139				289	337				
40	90	140				290					
41	91	141				291					
42	92	142				292					
43	93	143				293					
44	94	144				294					
45	95	145				295					
46	96	146				296					
47	97	147				297					
48	98	148				298					
49	99	149									
50	100	150									

